RekindledFX — Lava

Procedural lava flow shader + rising embers for Unity URP

Version 1.0.0

1. Requirements & Setup

- **Render pipeline:** Universal Render Pipeline (URP). This package is **URP only** — it does not support the Built-in Render Pipeline or HDRP in this version.
- **Unity version:** Developed and tested on Unity 6 (URP).
- **Bloom (recommended):** The lava's heat glow is HDR emissive. For the intended look, add a **Global Volume** with **Bloom** enabled. Everything still works without it, but the glow is much stronger with Bloom on.
- **No setup steps, no dependencies.** All textures are generated procedurally at runtime. Just import the package and use the prefabs or material.

2. Quick Start (60 seconds)

1. Open `Assets/RekindledFX/Elemental/Demo/LavaDemo.unity`.
2. Look at the Scene view — the lava previews live in edit mode. Press **Play** to see it running.
3. To use it in your own scene, either:
   - Drag the `LavaPool` prefab (`Assets/RekindledFX/Elemental/Prefabs/`) into your scene, or
   - Apply the `Lava` material (`Assets/RekindledFX/Elemental/Materials/`) to any mesh, then add a `LavaEmbers` prefab/component above it.

3. What's Included

Item	Path	Description
LavaFlow shader	`Shaders/LavaFlow.shader`	The animated lava surface shader (`RekindledFX/Elemental/LavaFlow`).
ElementalParticle shader	`Shaders/ElementalParticle.shader`	Additive particle shader used by the embers.
LavaEmbers component	`Runtime/LavaEmbers.cs`	Rising ember particles, configured entirely from code.
Materials	`Materials/`	`Lava` and `LavaCurrent` — ready-to-use presets of the LavaFlow shader.
Prefabs	`Prefabs/`	`LavaPool` (surface + embers) and `LavaEmbers`.
Demo scene	`Demo/LavaDemo.unity`	A minimal scene showing the effect.

4. Component Reference — LavaEmbers

Add this component to a GameObject placed above your lava surface (its local XZ plane sits on the surface); embers rise upward. It builds its ParticleSystem entirely from code and previews in edit mode.

Field	Description
Area	Rectangular area (X, Z in meters) that embers spawn from.
Rate	Embers produced per second.
Size	Min/max particle size.
Lifetime	Min/max particle lifetime (seconds).
Rise Speed	Upward drift speed.
Floatiness	Amount of organic wobble (Noise).
Start Color	Ember color at birth (HDR — hot orange by default).
End Color	Color it cools to at end of life (HDR).
Particle Shader	Reference to the ElementalParticle shader. Assigned in the prefab so the shader is included in player builds. Leave assigned — see the note in section 7.

5. Shader Reference — RekindledFX/Elemental/LavaFlow

Apply this shader (or the included `Lava` material) to any mesh. It is fully procedural and animated.

Lava Colors

- `Lava Hot Color`, `Lava Mid Color`, `Lava Cold Color` — the three-tone gradient of the molten surface.
- `Crust Color` — the color of the cooled crust between flows.

Flow

- `Flow Speed` — how fast the lava scrolls.
- `Flow Direction (XY)` — flow direction vector.
- `Flow Scale` — size of the flow pattern.
- `Turbulence` — swirl/distortion in the flow.

Bubbles

- `Bubble Scale`, `Bubble Speed`, `Bubble Intensity` — molten surface bubbling.

Heat Glow (Emission)

- `Emission Intensity` — overall glow strength (HDR; needs Bloom to shine).
- `Glow Pulse Speed`, `Glow Pulse Amount` — pulsing of the glow.
- `Emission Tint (glow color)` — tints the glow (HDR). Great for recolors.
- `Hot White Boost` — controls how much the hottest areas blow out toward white (lower = more color-preserving).

Surface

- `Roughness`, `Bump Strength` — surface lighting response.
- `Crust Threshold` — where crust forms vs. molten lava.

Vertex Displacement

- `Displacement Amplitude`, `Displacement Speed` — subtle surface movement (bulging).

Crust Plates and Cracks (optional — start with Crust Plate Amount = 0)

- `Crust Plate Amount (0=off)`, `Crust Plate Scale`, `Crust Plate Threshold` — hardened plate look.
- `Crack Width`, `Crust Flow Speed Mul`, `Crack Glow` — glowing cracks between plates.

Edge Heat Glow (optional)

- `Edge Glow (0=off)`, `Edge Glow Power` — fresnel rim heat glow on silhouettes.

Bubble Pop (optional)

- `Bubble Pop (0=off)` — occasional bursting bubbles.

Mapping

- `World-space Flow (seamless tiling)` — enable for large surfaces to avoid UV stretching/seams.
- `World Scale` — texel scale when World-space Flow is on.

6. Recipes

Lava on any surface

Create a material using `RekindledFX/Elemental/LavaFlow` (or duplicate the `Lava` material), then assign it to your mesh's Mesh Renderer.

Large lava lakes without stretching

Enable **World-space Flow** on the material and tune **World Scale**. This keeps the pattern seamless across big planes and terrains.

Add rising embers

Add the `LavaEmbers` prefab (or component) above the surface. Match its **Area** to the size of your lava, and tint **Start/End Color** to the mood you want.

Recolor to toxic / alien lava

Change `Lava Hot/Mid/Cold Color`, `Crust Color`, and `Emission Tint`. For a toxic green look, shift all colors toward green and set a green emission tint; lower `Hot White Boost` to keep the color saturated.

Calmer / more violent lava

Tune `Flow Speed`, `Turbulence`, `Bubble Intensity`, and `Bubble Pop`. Turn on `Crust Plate Amount` and `Crack Glow` for a crusted, cracked-earth look.

7. Troubleshooting / FAQ

The lava material is pink / magenta.
Your project is not using URP, or URP is not active. This package is URP only. Switch your project to the Universal Render Pipeline.

The lava doesn't glow.
Add a Global Volume with **Bloom** enabled, and/or raise `Emission Intensity` on the material. The glow is HDR emissive and relies on Bloom for the bright look.

Embers appear pink or missing in a built player (but work in the Editor).
Make sure the **Particle Shader** field on the `LavaEmbers` component is assigned to the ElementalParticle shader (it is pre-assigned in the included prefabs). This guarantees the shader is included in your build. If you build from a hand-made GameObject, assign it there too.

The pattern stretches on a large plane.
Enable **World-space Flow** on the material and adjust **World Scale**.

Embers don't move in edit mode.
They preview via live simulation in the Scene view. If you ever see them paused entering Play mode, they resume automatically.

8. Support & More from RekindledFX

Support: be.snake@gmail.com — questions, bug reports, and feature requests are welcome.

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All built the same way: procedural, zero dependencies, URP.

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